

18. Implement Perceptron based IRIS classification

Aim

To implement **Perceptron-based Iris classification** using Python.

Software Required

- Python 3.x
- Scikit-learn
- NumPy

Procedure

1. Load the Iris dataset.
2. Select two classes for binary classification.
3. Split the data into training and testing sets.
4. Create a Perceptron model.
5. Train the model.
6. Predict the test data.
7. Calculate accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target

# Select Setosa and Versicolor
X = X[y != 2]
y = y[y != 2]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
)  
model = Perceptron(random_state=42)  
model.fit(X_train, y_train)  
y_pred = model.predict(X_test)  
print("Predicted Classes:")  
print(y_pred)  
print("\nAccuracy:",  
      accuracy_score(y_test, y_pred))
```

Output:

```
Predicted Classes:  
[1 1 1 0 0 0 0 1 0 0 0 0 1 0 1 0 1 1 0 0]  
  
Accuracy: 1.0
```

Result

Thus, **Perceptron-based Iris classification** was successfully implemented.